

Ana Lorena Ortiz Loyola

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Data science student (Tecnológico de Monterrey, expected 2027) with a security-focused academic background - lattice-based cryptography research and coursework in cryptography & security - paired with hands-on experience in data governance, risk, and compliance controls at Deutsche Bundesbank (AnaCredit), a highly regulated financial-services environment. Comfortable documenting security processes and controls clearly, and motivated to support ISMS governance, audits, and incident-response exercises.

Core Competencies

- **Cryptography & Security:** Lattice-based cryptography research (LWE, NTRU) and academic coursework in cryptography and security fundamentals
- **Governance, Risk & Compliance:** Rule-based validation and control design at a central bank (Deutsche Bundesbank), a highly regulated financial-services environment
- **Security Documentation:** Clear, structured documentation of technical processes, controls, and findings for both technical and non-technical stakeholders
- **Programming:** Python (Pandas, NumPy, Scikit-Learn), SQL; building and testing data pipelines
- **Systematic Testing & Auditing:** Automated migration and validation tests (row counts, referential integrity, aggregate consistency), systematic anomaly detection
- **Version Control & Agile Workflows:** Git, Jira, structured technical documentation

Professional Experience

- **Deutsche Bundesbank (AnaCredit)** **Frankfurt am Main, Germany**
Data Engineer / Intern 04/2026–07/2026
 - Developed automated migration tests (row counts, referential integrity, aggregate consistency) using Pandas DataFrames, verifying that new tooling reproduced identical business logic before roll-out
 - Implemented automated error detection identifying deviations (duplicate IDs, value-range violations) in regulatory reporting data - a rule-based control design and monitoring discipline directly applicable to ISMS governance and audits
 - Documented control processes and findings clearly for both technical and non-technical stakeholders, tracked via Jira and Git
 - Significantly reduced response time to data quality issues through end-to-end ownership, from detection to resolution

Projects

- **Tecnológico de Monterrey**
Lattice-Based Cryptography Research 10/2024–02/2025
 - Implemented and analyzed the complexity of lattice-based cryptographic algorithms (LWE, NTRU) in Python
 - Contributed to a scientific publication analyzing security parameters and runtime behavior across lattice dimensions

- **Personal Project (GitHub)**
Banking Data Pipeline – Data Cleaning & Audit Log 05/2026
 - Built an end-to-end pipeline for simulated bank transactions (150 accounts) with a normalized SQLite database and foreign-key integrity checks
 - Staged data cleaning with a fully auditable log; automated balance maintenance via SQL triggers; built using **Claude Code** as an AI coding assistant to accelerate implementation
- **Tecnológico de Monterrey**
Predictive Modeling – Healthcare Optimization 10/2024–12/2024
 - Analyzed 25+ years of Mexican national health data to identify patient risk profiles and resource bottlenecks
 - Built predictive models (Random Forest, Neural Networks) and K-Means clustering to segment high-risk groups, directly analogous to clustering structured records into meaningful categories

Education

- **Tecnológico de Monterrey** **Mexico**
BSc in Data Science 2022–2027
 Grade: 1.3 (Academic Excellence Scholarship; DAAD KOSPIE Scholarship). Topics: data analysis & statistics, machine learning & AI, cryptography & security, NLP, numerical optimization.
- **Universität Göttingen** **Germany**
Exchange Semester 2025–2026
- **Tecnológico de Monterrey** **Mexico**
Abitur 2019–2022
 Grade: 1.0. National Physics Olympiad 2020; Honorable Mention, Physics Olympiad 2021.

Languages

- Spanish (native), English (C1, TOEFL iBT 105/120), German (C1).

Publications

- Contributed to a scientific publication on lattice-based cryptography (LWE, NTRU security parameter and runtime analysis), with Prof. A. F. De Abiega L'Eglise, Tecnológico de Monterrey (citation pending publication).

Honors and Awards

- **Academic Excellence Scholarship** – Tecnológico de Monterrey.
- **DAAD KOSPIE Scholarship**.
- **National Physics Olympiad** – Mexico (2020).
- **Honorable Mention, Physics Olympiad** (2021).

References

- Available upon request.